HWA CHONG INSTITUTION

C2 PRELIMINARY EXAMINATION

9746 H2 CHEMISTRY

PAPER 1 Multiple Choice

29 September 2009 **1 hour**

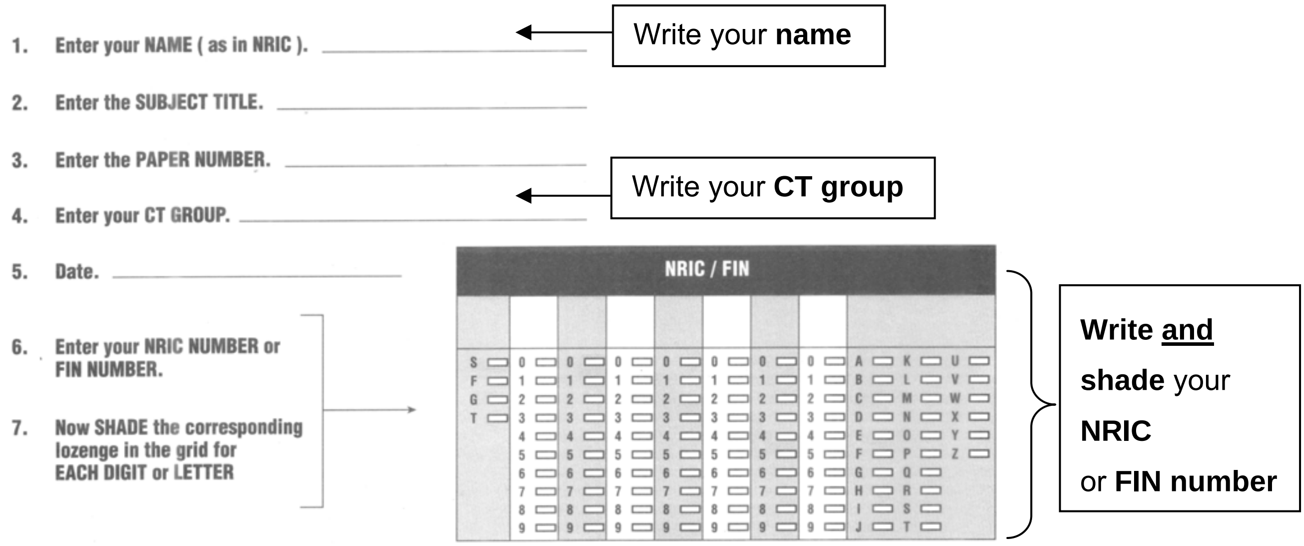
Do not open this booklet until you are told to do so.

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Complete the information on the optical mark sheet (OMS) as shown below.



There are **forty** questions on this paper. Answer **all** questions. For each question, there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

Each correct answer will score one mark. A mark will **not** be deducted for a wrong answer.

Any rough working should be done in this booklet.

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1 The relative abundances of the isotopes of a sample of titanium are shown in the table below.

Relative Isotopic Mass	46	47	48	49	50
Relative Abundance	11.2	10.1	100.0	7.3	7.0

What is the relative atomic mass of titanium in this sample?

A 47.92 **B** 47.91 **C** 47.90 **D** 47.89

2 In the middle of the 15th century, 'black powder' was widely used by the Arabs as propellant for muzzle-loading cannons. 'Black powder' consisted of a mixture of potassium nitrate, sulfur and charcoal (carbon) in the molar ratio shown in the equation below.

$$2KNO_3\,(s) + S\,(s) + 3C\,(s) \rightarrow K_2S\,(s) + N_2\,(g) + 3CO_2\,(g)$$

Given that 2.00 mol of gas could just expel the cannonball out of the cannon tube, what is the minimum mass of 'black powder' (in g), required for this to happen?
[M_r: KNO_3, 101.1]

A 101 **B** 135 **C** 202 **D** 270

3 The successive ionisation energies (IE) of two elements, **Q** and **R**, are given below.

IE / kJ mol^{-1}	1st	2nd	3rd	4th	5th	6th	7th	8th
Q	1090	2350	4610	6220	37800	47000	-	-
R	1251	2298	3822	5158	6542	9362	11018	33604

What is the likely formula of the compound that is formed when **Q** reacts with **R**?

A **QR** **B** $\mathbf{Q_2R_3}$ **C** $\mathbf{QR_4}$ **D** $\mathbf{Q_4R}$

4 Which of the following statements about the properties associated with ionic and covalent bonds is correct?

A Some covalent compounds can serve as electrolyte in water.
B Ionic bonds and covalent bonds cannot both occur in the same compound.
C Ionic compounds and metals can conduct electricity in both the solid and liquid states.
D Any covalent compounds that have both hydrogen and oxygen in its molecule can form intermolecular hydrogen bond.

5 A sample of 0.25 g of an organic compound is vaporised in a gas syringe and occupies 260 cm^3 at 50 °C and 0.5 atm.

What is the relative molecular mass of the compound?

A $\dfrac{0.25 \times 0.5 \times 260 \times 323}{24000 \times 298}$

B $\dfrac{0.25 \times 0.5 \times 260 \times 298}{24000 \times 323}$

C $\dfrac{0.25 \times 24000 \times 50}{0.5 \times 260 \times 25}$

D $\dfrac{0.25 \times 24000 \times 323}{0.5 \times 260 \times 298}$

6 The explosive nitroglycerin ($C_3H_5N_3O_9$) decomposes rapidly upon ignition or sudden impact according to the following equation:

$4C_3H_5N_3O_9(l) \rightarrow 12CO_2(g) + 10H_2O(g) + 6N_2(g) + O_2(g)$ $\Delta H^{\ominus}_r = -5678$ kJ mol^{-1}

Given the following data, what is the standard enthalpy change of formation of nitroglycerin in kJ mol^{-1}?

Standard enthalpy change of formation of $CO_2(g) = -394$ kJ mol^{-1}
Standard enthalpy change of formation of $H_2O(g) = -242$ kJ mol^{-1}

A −368
B −1470
C −3207
D +5042

7 The bromination of propanone is acid-catalysed.

$$CH_3COCH_3 + Br_2 \xrightarrow{H^+} CH_3COCH_2Br + H^+ + Br^-$$

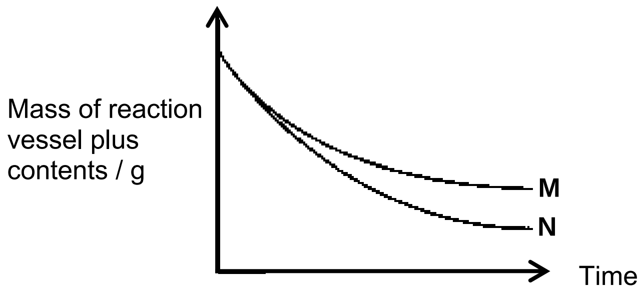
The rate of disappearance of the bromine colour was measured for several different concentrations of propanone, bromine and H^+ at a certain temperature and the results tabulated below.

Experiment	$[CH_3COCH_3]$/ mol dm^{-3}	$[Br_2]$/ mol dm^{-3}	$[H^+]$/ mol dm^{-3}	Rate of disappearance of Br_2 colour/ mol dm^{-3} s^{-1}
1	0.30	0.05	0.05	5.70×10^{-5}
2	0.30	0.10	0.05	5.70×10^{-5}
3	0.30	0.05	0.10	1.14×10^{-4}
4	0.40	0.05	0.20	3.04×10^{-4}

Which of the following statements about the above reaction is true?

A The rate equation for the reaction is rate = $k[CH_3COCH_3][Br_2]$.
B The rate constant for the reaction is 3.8×10^{-3} mol^{-1} dm^3 s^{-1}.
C The rate constant of the reaction doubles when $[CH_3COCH_3]$ is doubled.
D The rate constant of the reaction remains unchanged when temperature is doubled.

8 Two experiments, **M** and **N**, were carried out to investigate the rates of the reaction between copper(II) carbonate and excess acid. The initial mass of the reaction vessel plus contents was the same in both experiments. The results are shown in the diagram below.



Which of the following change in the conditions from Experiment **M** to Experiment **N** might explain the results shown?

A increasing the concentration of the acid
B increasing the mass of copper(II) carbonate
C decreasing the particle size of the copper(II) carbonate
D adding a catalyst

9 Phosgene, $COCl_2$, is a poisonous gas used as a chemical weapon in World War I. It can dissociate into two other poisonous gases, carbon monoxide and chlorine, at high temperatures according to the equilibrium below:

$$COCl_2\,(g) \rightleftharpoons CO\,(g) + Cl_2\,(g)$$

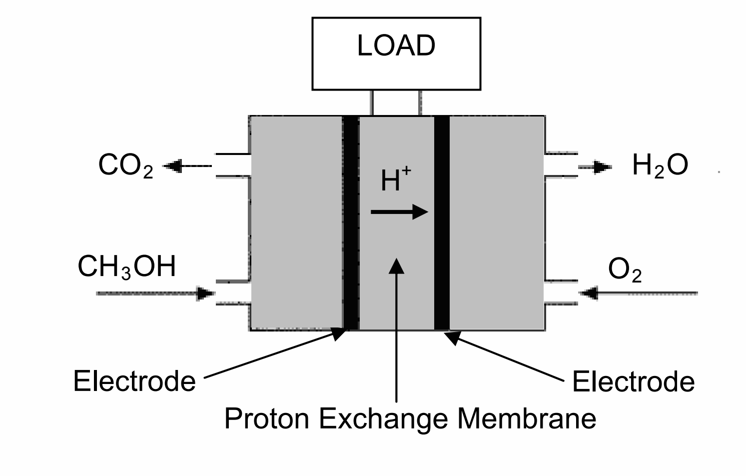
When 1 mol of phosgene is heated in a closed vessel at 600 K, the final pressure at equilibrium was 5 atm. Given that the mole ratio of phosgene to chlorine at equilibrium is 15:2, what is the numerical value of K_p at 600 K?

A 0.0140 **B** 0.0157 **C** 0.0702 **D** 0.0784

10 A solution of an acid **E** has the same pH as a solution of acid **F.** Equal dilution increases the pH of acid **E** more than that of acid **F**. Which of the following pairs of acids would show this behaviour?

	E	**F**
A	H_3PO_4	HCl
B	HCl	CH_3CO_2H
C	HCl	H_2SO_4
D	CH_3CO_2H	H_2SO_4

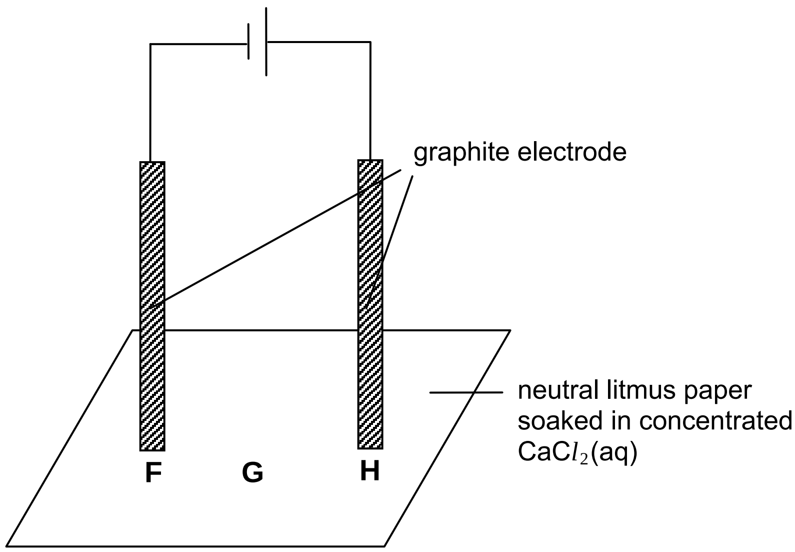
11 Direct-methanol fuel cells make use of methanol as the fuel to generate energy. They are limited in the power they can produce, but can still store high energy content in a small space. The electrolyte in this fuel cell is a dilute aqueous acid.



Which of the following statements is **incorrect**?

A Methanol is oxidized in this reaction.
B The proton exchange membrane functions as a salt bridge.
C The pH at the anode increases temporarily as energy is generated.
D Electrons flow from the methanol/carbon dioxide half-cell to that of the oxygen/ water half-cell.

12 A direct current is passed through the apparatus shown in the diagram below.

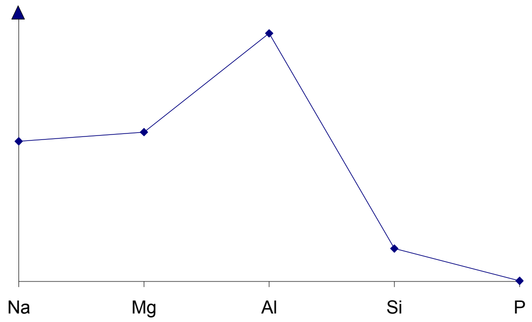


After a few minutes, what colours would be observed on the paper at the three points **F**, **G** and **H**?

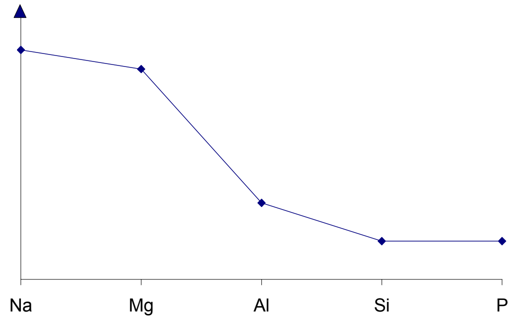
	F	G	H
A	red	white	blue
B	blue	purple	white
C	blue	purple	red
D	purple	purple	white

13 The graphs below show the variation in two properties of the elements Na to P and their compounds.

Graph I



Graph II



Which properties are illustrated in Graphs I and II?

	Graph I	Graph II
A	electrical conductivity of the element	pH of the chloride when added to water
B	electrical conductivity of the element	pH of the oxide when added to water
C	melting point of the element	pH of the chloride when added to water
D	melting point of the element	pH of the oxide when added to water

14 **X**, **Y** and **Z** are elements in Period 3 of the Periodic Table.

A mixture containing the oxides of **X**, **Y** and **Z** was dissolved in excess dilute sulfuric acid and filtered. The oxide of **Z** was collected as a residue. When excess dilute sodium hydroxide was added to the filtrate, only a white precipitate of the hydroxide of **Y** was formed.

What are the possible identities of **X**, **Y** and **Z**?

	X	**Y**	**Z**
A	Mg	A*l*	P
B	A*l*	Mg	P
C	Mg	A*l*	Si
D	A*l*	Mg	Si

15 The reactivity of the Group II metals with oxygen increases down the group.

Which statement best explains this trend?

- **A** Electronegativity of the elements decreases down the group.
- **B** Ionic radius of the metal cations increases down the group.
- **C** Lattice energy of the metal oxides decreases down the group.
- **D** Reducing power of the elements increases down the group.

16 A student was given a sample of calcium carbonate that was mixed with calcium nitrate. He carried out two separate experiments using the same mass of the solid mixture.

Experiment 1: The sample was heated strongly to constant mass and the volume of gas evolved was found to be 200 cm^3

Experiment 2: Excess dilute sulfuric acid was added to the sample and volume of gas evolved was found to be 75 cm^3.

All volumes were measured at the same temperature and pressure.

What is the mole ratio of calcium carbonate to calcium nitrate in the sample?

A 3 : 2
B 3 : 5
C 6 : 5
D 9 : 5

17 0.03 mol of chlorine gas was bubbled into 100 cm^3 of hot aqueous sodium hydroxide of concentration 1 mol dm^{-3}.

Which of the following statement is correct regarding the above reaction?

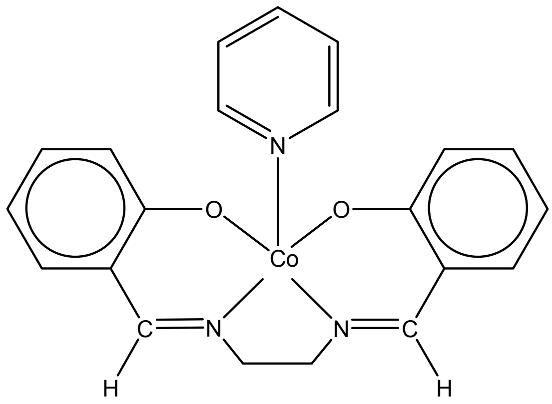
A 0.03 mol of sodium chlorate (I) was formed.
B 0.03 mol of sodium chlorate (V) was formed.
C The chloride produced required 0.05 mol of silver nitrate for complete precipitation.
D The excess sodium hydroxide required 0.06 mol of dilute hydrochloric acid for complete neutralization.

18 The solubility of the silver halides in aqueous ammonia decreases from $AgCl$ to AgI.

What helps to explain this trend?

A As a more powerful ligand, NH_3 can displace Cl^- ions and Br^- ions, but not I^- ions.
B Cl^- ions and Br^- ions form complexes with NH_3(aq), but I^- ions do not.
C The value of the solubility product of the silver halides decreases from $AgCl$ to AgI.
D The covalent bonding between Ag and the halogen atom increases in strength from $AgCl$ to AgI.

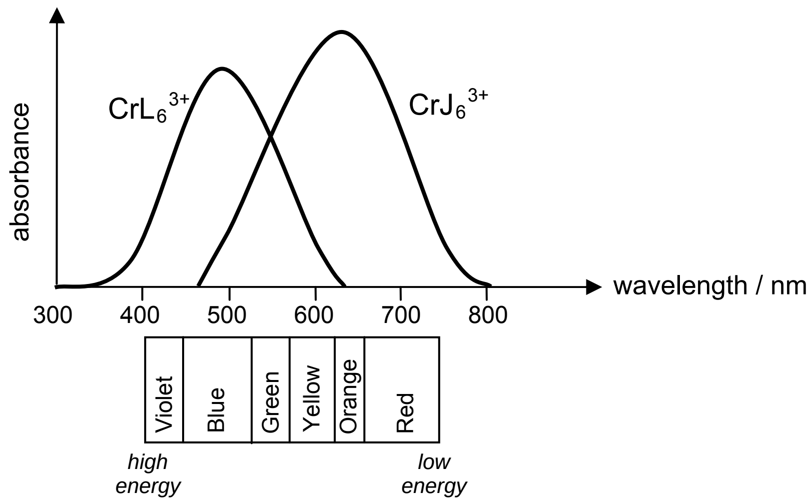
19 The following cobalt complex is known to be the functional model for biological oxygen carrier.



What is the electronic configuration of the cobalt cation in the above complex?

A [Ar]$3d^6$ **B** [Ar]$3d^7$ **C** [Ar]$3d^5 4s^2$ **D** [Ar]$3d^7 4s^2$

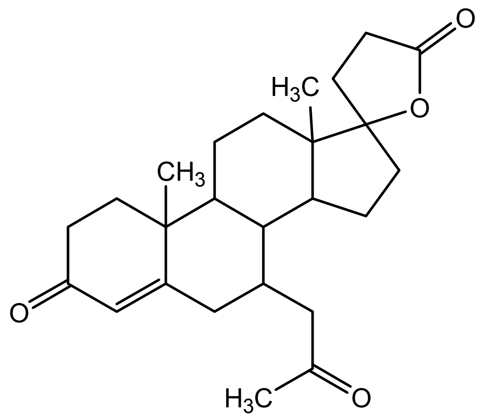
20 The diagram below shows the visible spectra of two chromium (III) complexes, CrL_6^{3+} and CrJ_6^{3+}. The various colours corresponding to the approximate wavelengths in the visible light region are shown below the axis.



What is the colour of each complex and which ligand causes a larger d-orbital splitting?

	colour of CrL_6^{3+}	*colour of CrJ_6^{3+}*	*ligand which causes a larger d-orbital splitting*
A	blue	yellow orange	L
B	blue	yellow orange	J
C	red	violet	L
D	red	violet	J

21 Spironolactone is an anti-androgen used as a common component in hormone therapy.

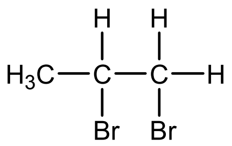


How many stereoisomers does spironolactone have?

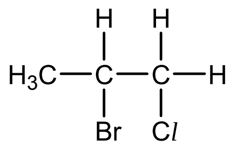
A 2^3 **B** 2^4 **C** 2^6 **D** 2^7

22 Which of the following compounds will **not** be produced when propene reacts with a mixture of Br_2 in aqueous Na*Cl*?

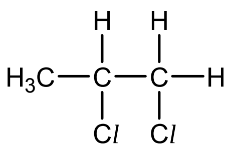
A



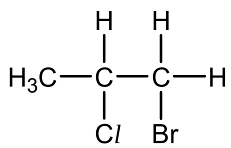
C



B



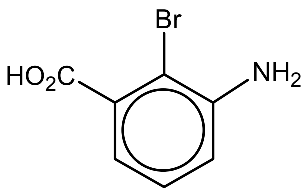
D



23 During electrophilic substitution on a benzene ring, the position of the incoming group, the new substituent, is determined by the nature of the group(s) already present in the ring. The directing effects of these groups are shown in the table below.

Directing Effects	3 – directing	2,4 – directing
Functional Groups with these effects	$-NO_2$, $-CO_2H$,	$-CH_3$, $-NH_2$, –X (halogens)

Compound **V** may be synthesised from benzene.

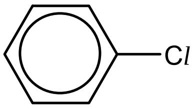
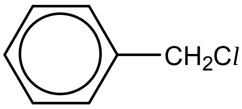
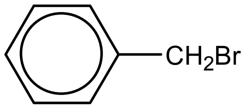
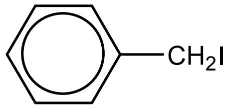


V

Which of the following synthetic routes will yield compound **V**?

A Nitration → alkylation → reduction → bromination → oxidation
B Alkylation → nitration → oxidation → bromination → reduction
C Bromination → nitration → reduction → alkylation → oxidation
D Nitration → reduction → bromination → alkylation → oxidation

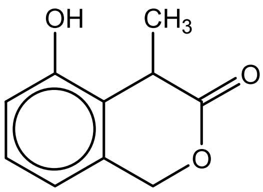
24 The table below shows the rate of hydrolysis of the halogen-containing compounds **P** to **S**. The rate of hydrolysis is measured by the speed at which the silver halide precipitate is formed.

P	**Q**	**R**	**S**
			
No precipitate formed with prolonged heating with ethanolic $AgNO_3$.	Precipitate formed after 10 minutes of heating reaction mixture with ethanolic $AgNO_3$.	Precipitate formed after 2 minutes of warming reaction mixture with ethanolic $AgNO_3$.	Precipitate forms instantaneously with ethanolic $AgNO_3$.

Which statement explains the rate of reaction?

A The C–X bond length increases from compound **P** to **S**.
B The solubility of the compounds increases from **P** to **S**.
C The mechanism for the reaction changes from **P** to **S**.
D The reaction conditions are increasingly milder from **P** to **S**.

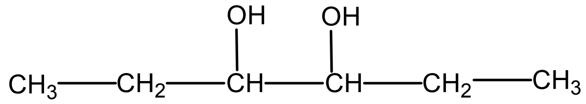
25 Which of the following deductions about the reactions of **G** can be made from its structure?

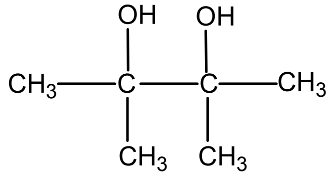


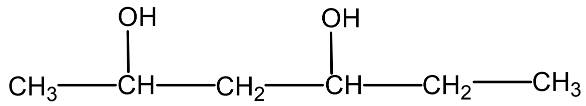
G

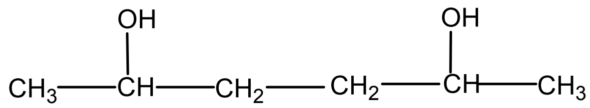
A It gives white fumes of HC*l* with PC*l*$_5$.
B It gives a yellow precipitate with aqueous alkaline iodine.
C It gives an orange precipitate with 2,4-dinitrophenylhydrazine.
D It gives a violet colouration with neutral FeC*l*$_3$ solution.

26 Which of the isomers of $C_6H_{14}O_2$ gives, on complete dehydration, the greatest number of isomers, including stereoisomers?

A 

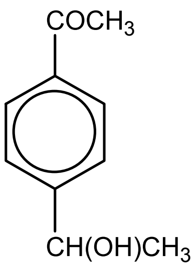
B 

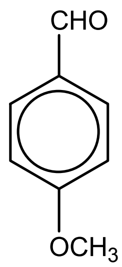
C 

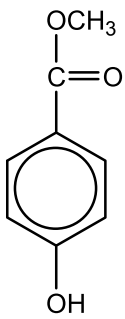
D 

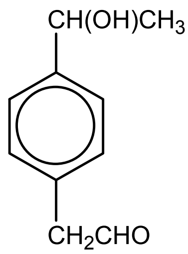
27 A compound **U** has all the following properties:
- It is neutral.
- It gives a silver mirror with Tollens' reagent.
- It decolourises a hot aqueous solution of acidified $KMnO_4$ with evolution of CO_2 gas.

What could **U** be?

A 

B 

C 

D 

28 Which reaction yields a carbon compound incorporating deuterium, D? [D = ^{2}H]

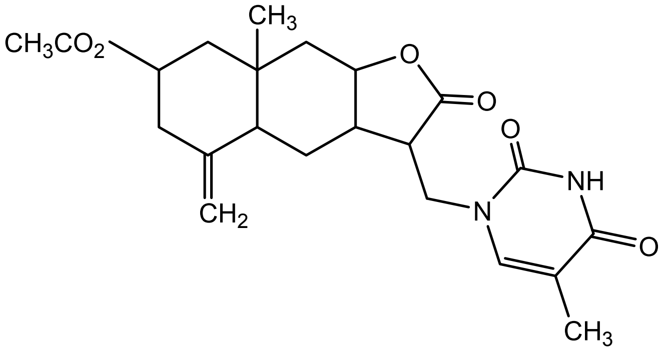
A $CH_3CH_2CN \xrightarrow[D_2O]{NaOD}$

B $CH_3CD_2OH \xrightarrow[\text{heat}]{\text{acidified } KMnO_4}$

C $CH_3COCH_3 \xrightarrow[\text{dry ether}]{LiAlD_4}$

D $(CH_3)_3COH \xrightarrow[\text{heat}]{\text{conc. } D_2SO_4}$

29 The sesquiterpene lactone, **M**, extracted from sunflowers, was studied to determine its anti-tumour potency.

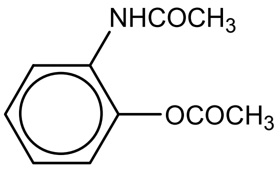


M

How many moles of dilute NaOH are required to completely hydrolyse 1 mole of **M**?

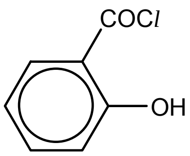
A 3 **B** 4 **C** 5 **D** 6

30 A student attempted to synthesize the following compound **N**.

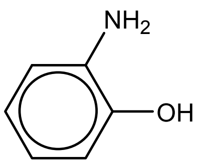


N

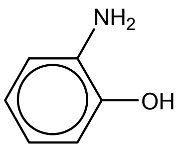
Which of the following pairs of reactants will give the target molecule?

A 

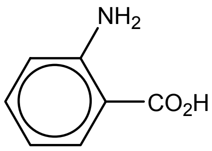
and CH_3NH_2

B 

and CH_3COCl

C 

and CH_3CO_2H

D 

and CH_3COCl

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct. (You may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

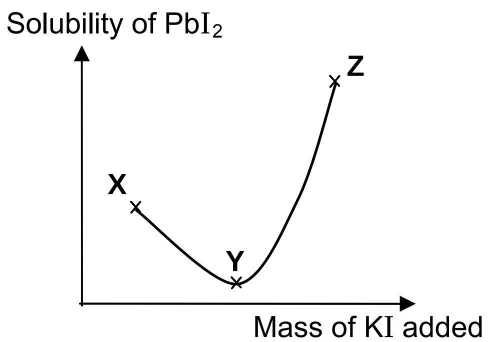
31 Which of the following statements are true about $NaHF_2$?

1 $NaHF_2$ conducts electricity in the molten state.
2 The HF_2^- ion contains two covalent bonds.
3 The shape of the anion is bent.

32 Which of the following have the same value as the standard enthalpy change of formation of carbon monoxide?

1 $\frac{1}{2}\,\Delta H_f^\ominus(CO_2)$
2 $\Delta H_f^\ominus(CO_2) - \Delta H_c^\ominus(CO)$
3 $\Delta H_c^\ominus(C) - \Delta H_c^\ominus(CO)$

33 The following graph represents how the solubility of a sparingly soluble salt lead(II) iodide, PbI_2, changes upon addition of solid potassium iodide.



Which of the following statements are correct?

1 The change in solubility along **XY** is caused by a common ion effect.
2 The change in solubility along **YZ** is caused by the formation of a complex between Pb^{2+} and I^- ions.
3 At **Y**, the molar concentration of I^- ions is twice that of Pb^{2+} ions.

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

34 Which of the following could act as acidic buffers?

1 A 1:2 mixture of HCl and CH_3CO_2Na
2 A 1:2 mixture of $NaHCO_3$ and Na_2CO_3
3 A 1:2 mixture of NaOH and NH_4Cl

35 Astatine is below iodine in Group VII of the Periodic Table.

Which of the following properties are likely to be true for astatine and its compounds?

1 When a heated glass rod is inserted into a container of hydrogen astatide, it decomposes into its elements.
2 Astatine is a weaker oxidizing agent than iodine.
3 Silver astatide has more covalent character than silver iodide.

36 *Use of the Data Booklet is relevant to this question.*

The colours of various vanadium ions are given in the table below.

Oxidation state	V	IV	III	II
Colour	yellow	blue	green	violet

When an excess of gas **W** was bubbled into an aqueous solution of acidified ammonium vanadate(V), NH_4VO_3, the solution turns green.

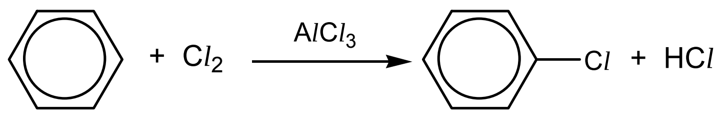
Which of the following substances could be gas **W**?

1 Hydrogen
2 Sulfur dioxide
3 Nitrogen dioxide

37 In which of the following reactions is the reactive carbon tetrahedral in the organic intermediate?

1 $CH_4 + Cl_2 \xrightarrow{uv} CH_3Cl + HCl$

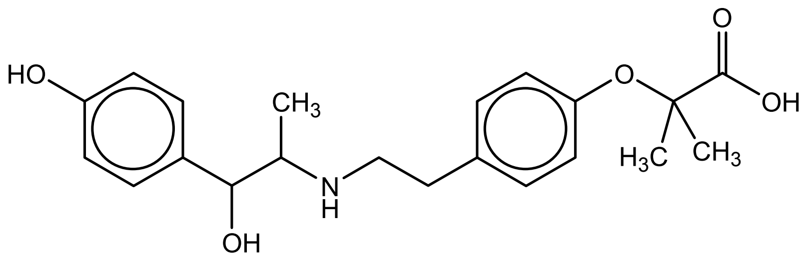
2 $HCHO + HCN \xrightarrow{\text{trace KCN}} CH_2(OH)CN$

3 

38 Which statements about the complete combustion of an alkene, C_nH_{2n}, in oxygen are correct?

1 The volume of oxygen required is directly proportional to the number of carbon atoms present in the molecule.
2 The volume of gas produced at 25 °C is the same as for the complete combustion of an alkane with the same number of carbon atoms per molecule.
3 At 120 °C, the volume of steam produced is always twice the volume of carbon dioxide.

39 Compound **F** is used to treat urinary incontinence. The structure of **F** is shown below.



F

Which of the following statements about the reactions of **F** are **incorrect**?

1 1 mol of **F** reacts with $SOCl_2$ to release 3 mol of HCl gas.
2 1 mol of **F** reacts with Na metal to release 3 mol of H_2 gas.
3 1 mol of **F** reacts with Na_2CO_3 to release 2 mol of CO_2 gas.

40 Which sequence shows the nitrogen compounds in increasing order of pK_b value?

1 $CH_3CH_2NHCH_3$, $C_6H_5CH_2NH_2$, CH_3CONH_2
2 $C_6H_5NH_2$, NH_3, CH_3NH_2
3 $CH_3CH_2N(CH_3)_2$, $C_6H_5NH_2$, $CH_3CH_2NHCH_3$

BLANK PAGE